CUSTOMER SUPPORT TICKET PRIORITY PREDICTION AND AUTOMATED ASSIGNMENT SYSTEM USING AGENTFORCE

ALPHA COLLEGE OF ENGINEERING

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USER STORY:

The **Customer Support Ticket Priority Prediction and Automated Assignment System** uses Salesforce CRM and Agentforce to automate support ticket handling.

When a support user inputs an **Account Name** into the **Support Ticket Priority Analysis** Agentforce subagent, an **Auto-Launched Flow** retrieves the Account's latest support ticket and evaluates its description against specific keywords:

- **High Priority:** Triggered by keywords like *"urgent"*, *"not working"*, or *"failure"*. Automates assignment to a senior support agent, creates a High-priority follow-up Task (*"Urgent Ticket Handling"*, status *Not Started*), and sends an immediate escalation message.
- **Medium Priority:** Triggered by keywords like *"issue"*, *"slow"*, or *"delay"* (when no High-priority condition applies).
- **Low Priority:** Default level assigned when no keywords match.

The system returns key details—**Ticket ID**, **Priority Level**, **Assigned Agent**, and **Action Message**—back to Agentforce for display in the conversation, eliminating manual triage and accelerating resolution times.

1. INTRODUCTION

1.1 Project Overview :

This project demonstrates a Salesforce-based system for analyzing customer support tickets and determining their priority. It uses a custom object to store ticket information, an Auto-Launched Flow to retrieve and classify the latest ticket for an Account, and Agentforce to provide a conversational way to run the analysis.
The Flow classifies tickets as High, Medium, or Low based on keywords in the ticket description. For a High-priority ticket, it creates an urgent-handling Task and returns an action message and assignment information.

1.2 Purpose

The project aims to:

- Store support ticket information in Salesforce.
- Retrieve the latest support ticket associated with an Account.
- Classify the ticket as High, Medium, or Low based on its description.
- Create an urgent-handling Task for a High-priority ticket.
- Provide ticket-analysis results through an Agentforce subagent.

2. IDEATHON PHASE

2.1 Problem Statement(Customer Problem Statement Template)

User	Need	Problem	Impact
Support Agent	Identify tickets that may need urgent attention	Ticket urgency may not be immediately clear from a description	Important tickets could be overlooked
Customer	Have critical issues recognized quickly	A support request may need review before its urgency is understood	The customer may be concerned about delays
Support Manager	Review ticket priority consistently	Assessing ticket descriptions manually can take time	Ticket handling may be inconsistent

2.2 Empathy Map Canvas

User persona: Support Agent

- **Thinks and feels**: Wants to understand ticket urgency quickly.
- **Sees**: Ticket details, issue type, description, status, and priority fields.
- **Says and does**: Reviews support requests and follows up on assigned work.
- **Pain points**: Manually scanning descriptions and identifying urgency signals.
- **Potential gains**: A consistent priority result and automatic Task creation for High-priority cases.

2.3 Brainstorming and Idea Selection

The project focuses on Salesforce because the completed implementation uses Salesforce records, a custom object, Flow automation, and an Agentforce subagent. The selected solution retrieves ticket information and applies configured keyword rules to classify priority.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey

The implemented process is:

Account Name provided → Account retrieved → Latest related ticket retrieved → Description analyzed → Priority determined → High-priority Task created, if applicable → Results returned

3.2 Requirements

Functional requirements

ID	REQUIREMENT
FR-1	Store support ticket information in the Support Ticket Intelligence custom object.
FR-2	Associate tickets with an Account and, optionally, a Contact.
FR-3	Retrieve the latest ticket associated with the supplied Account.
FR-4	Classify the ticket as High, Medium, or Low using configured description keywords.
FR-5	Create an urgent-handling Task for a High-priority ticket.
FR-6	Return the ticket ID, priority, assignment information, and action message.
FR-7	Allow the ticket-analysis process to be invoked through Agentforce.

Non-functional considerations

- **Usability:** The Agentforce interaction accepts an Account Name and returns a readable result.
- **Reliability:** Results depend on the configured Flow conditions and available Salesforce data.
- **Security:** Access to Accounts, tickets, and Tasks depends on Salesforce permissions.
- **Maintainability:** Keyword rules and Flow behavior can be updated through configuration.

3.3 Data Flow

User → Agentforce subagent → Auto-Launched Flow → Account and ticket records → Keyword-based classification → High-priority Task, if applicable → Response

3.4 Technology and Salesforce Components

COMPONENT	CONFIGURATION USED

Platform	Salesforce
Ticket storage	Support_Ticket_Intelligence__c
Automation	Auto-Launched Flow
Conversational interaction	Agentforce subagent
Related Salesforce records	Account, Contact, User, and Task

4. PROJECT DESIGN

4.1 Problem and Solution Fit

Problem	Implemented approach
Ticket descriptions need to be reviewed for urgency	Flow checks the description against configured keywords.
The latest ticket needs to be identified	Flow retrieves the latest ticket associated with the supplied Account.
High-priority cases need an urgent follow-up action	Flow creates an urgent-handling Task when the priority is High.
Users need a clear result	The Flow returns priority and action information for the Agentforce conversation.

4.2 Proposed Solution

The solution uses an Auto-Launched Flow to accept an Account Name, retrieve the Account and its latest support ticket, analyze the ticket description, and return the classification. For High-priority tickets, the Flow creates a Task with the subject **"Urgent Ticket Handling."** The Agentforce subagent invokes the process and presents the outputs in conversation.

4.3 Solution Architecture

User → Agentforce: Support Ticket Priority Analysis → Auto-Launched Flow → Account lookup → Latest ticket lookup → Priority decision → Task creation for High priority → Flow outputs → Agentforce response

5. PROJECT PLANNING AND SCHEDULING

The provided project material describes a multi-sprint planning template. However, the completed work described for this project is limited to the custom object, Auto-Launched

Flow, and Agentforce configuration. No completed sprint dates, story-point totals, velocity, or burndown results were provided, so these are not reported as completed project outcomes.
The work completed can be summarized as:

1. Configure the Support Ticket Intelligence custom object and fields.
2. Build and activate the Auto-Launched Flow.
3. Configure the Agentforce subagent and its Flow action.
4. Test the ticket-analysis interaction and outputs.

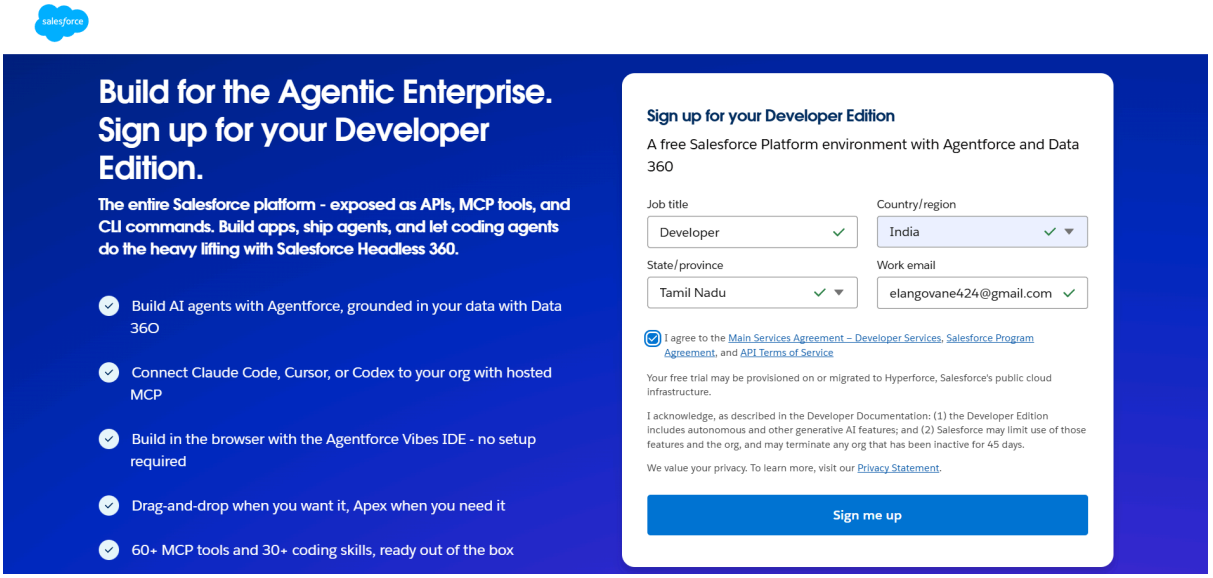
6. PROJECT DEVELOPMENT

Activity 1: Creating Developer Account

Creating a developer org in salesforce.

1. Go to https://developer.salesforce.com/signup

2. On the sign up form, enter the following details



1) First name & Last name

2) Email

3) Role : Developer

4) Company : College Name

5) County : India

6) Postal Code : pin code

7) Username : should be a combination of your name and company

This need not be an actual email id, you can give anything in the format :

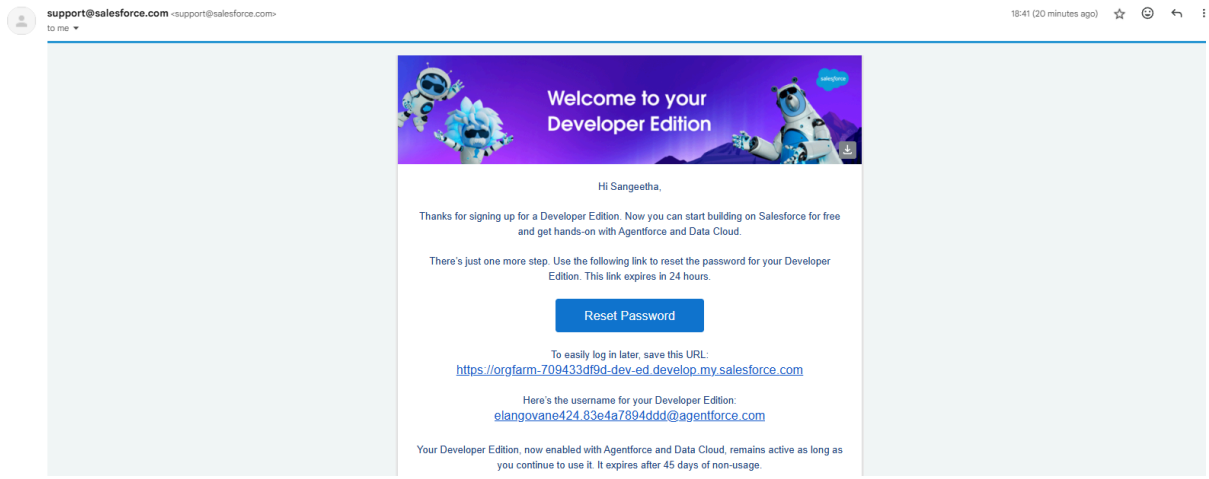
username@organization.com

Click on sign me up after filling these.

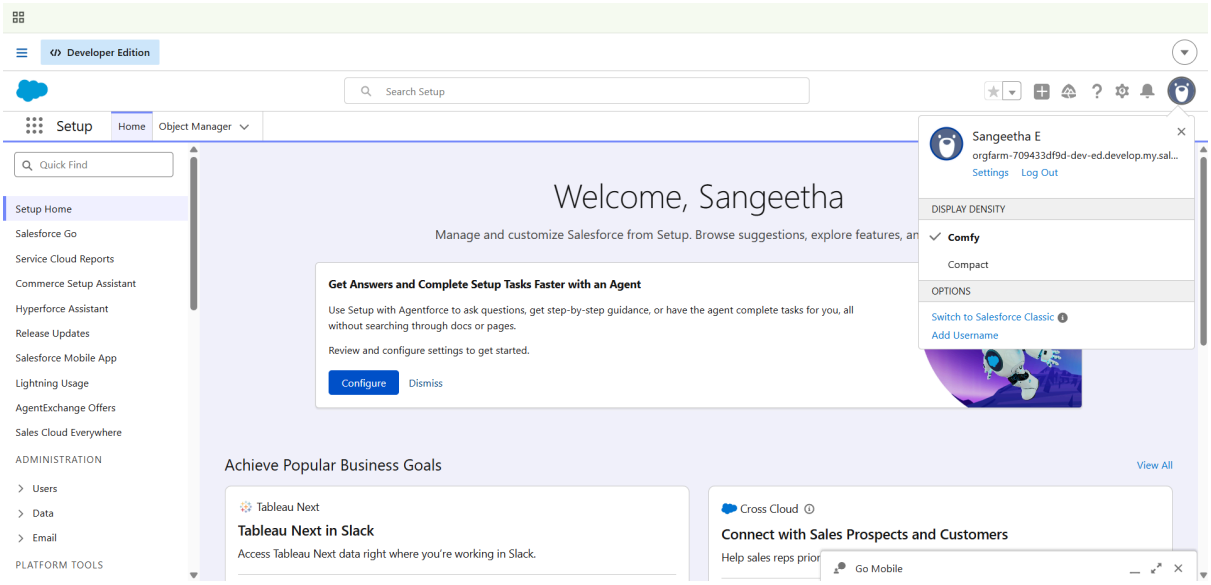
Activity 2: Account Activation

1. Go to the inbox of the email that you used while signing up. Click on the verify

account to activate your account. The email may take 5-10mins.



2. Click on Verify Account / Reset Password
3. Give a password and answer a security question and click on change password.
4. Then you will redirect to your salesforce setup page.



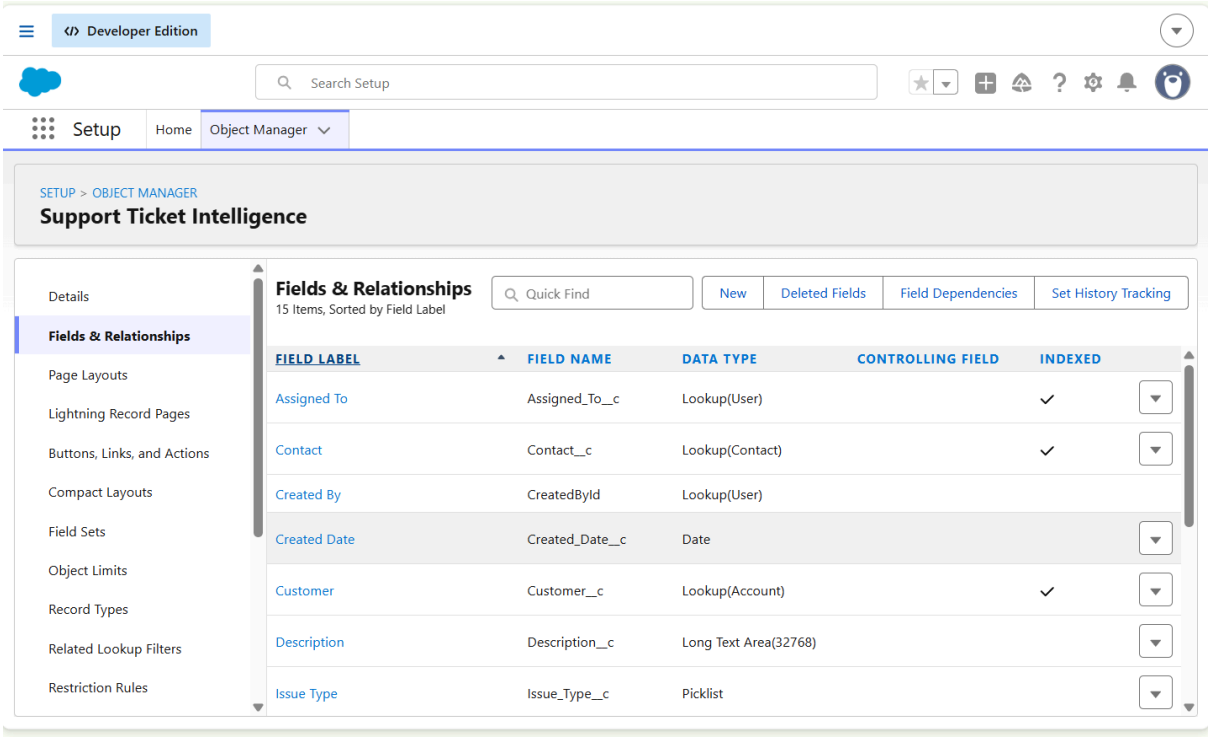
6.1 Custom Object

Object label: Support Ticket Intelligence

API name: Support_Ticket_Intelligence__c

FIELD LABEL	API NAME	DATA TYPE	DESCRIPTION
Ticket Number	Ticket_Number__c	Auto Number	TKT-{0000}
Customer	Customer__c	Lookup (Account)	Related Account

Contact	Contact__c	Lookup (Contact)	Customer contact
Issue Type	Issue_Type__c	Picklist	Technical, Billing, General
Description	Description__c	Long Text Area	Issue details
Priority Level	Priority_Level__c	Picklist	Low, Medium, High
Status	Status__c	Picklist	New, In Progress, Resolved
Created Date	Created_Date__c	Date	Ticket date
Assigned To	Assigned_To__c	Lookup (User)	Support agent
SLA Breach Risk	SLA_Breach_Risk__C	Checkbox	Risk flag
Resolution Time (hrs)	Resolution_Time__c	Number	Time taken



6.2 AUTO-LAUNCHED FLOW

Flow label: Support_Ticket_Intellegence

Flow type: Auto-Launched Flow

Purpose: Retrieve the latest ticket for an Account, classify its priority, create a Task for High-priority tickets, and return results to Agentforce.

Flow Logic

LABEL	API NAME	TYPE	DIRECTION
Account Name	varAccountName	Text	Input
Account Id	varAccountId	Text	Output
Ticket Id	varTicketId	Text	Output
Priority Level	varPriorityLevel	Text	Output
Assigned Agent	varAssignedTo	Text	Output
Action Message	varActionMessage	Text	Output

Flow logic

1. **Get Account:** Retrieve the Account whose Name matches varAccountName, storing only the first record.

   Element: Get Records
   - Label: Get Account
   - Object: Account
   - Condition:
     - Name Equals {!varAccountName}
   - Sort: CreatedDate DESC
   - Store: Only first record
2. **Store Account Id:** Set varAccountId to the retrieved Account's ID.

   Element: Get Records
   - Label: Get Account
   - Object: Account
   - Condition:
     - Name Equals {!varAccountName}
   - Sort: CreatedDate DESC
   - Store: Only first record
3. **Get Ticket:** Retrieve the latest Support_Ticket_Intelligence__c record whose Customer__c matches varAccountId.

   Element: Get Records
   - Label: Get Ticket

- Object: Support_Ticket_Intelligence__c
- Condition:
    - Customer__c Equals {!varAccountId}
- Sort: CreatedDate DESC
- Store: Only first record

4. **Store Ticket Id:** Set varTicketId to the retrieved ticket's ID.

   Element: Assignment

   - Label: Store Ticket Id

   Set:

   - varTicketId = {!Get_Ticket.Id}

5. **Analyze Description:** Check the ticket description against the configured keywords:
   - **High:** "urgent," "not working," or "failure"
   - **Medium:** "issue," "slow," or "delay"
   - **Low:** none of the listed keywords

   Element: Decision

   - Label: Analyze Description

   Outcomes:

   High Priority

   - Description contains:
     - "urgent"
     - "not working"
     - "failure"

(Use multiple conditions with CONTAINS)

Medium Priority

- Description contains:
  - "issue"
  - "slow"
  - "delay"

Default Outcome

- Low Priority

6. **Set Priority Level:** Set varPriorityLevel to High, Medium, or Low according to the matching outcome.

   High Path

   - varPriorityLevel = "High"

   Medium Path

- varPriorityLevel = "Medium"

Low Path

- varPriorityLevel = "Low"

7. **Check for High Priority:** If the priority is High, continue to the Task-creation step.

   Element: Decision

   - Label: Is High Priority

   Condition:

   - varPriorityLevel Equals "High"

8. **Create Task for High Priority:** Create a Task with:

   Element: Create Records

   - Object: Task

   Set Fields:

   - Subject = "Urgent Ticket Handling"
   - WhatId = {!varTicketId}
   - Priority = High
   - Status = Not Started

9. **Set Assignment Output:** Set varAssignedTo to **Senior Support Agent**, as specified in the Flow configuration.

   Element: Assignment

   - Label: Set Assigned Agent

   Set:

   - varAssignedTo = "Senior Support Agent"

10. **Optional SLA check:** The design notes include an optional check for tickets older than two days. This should be treated as optional logic, not as a confirmed implemented or tested feature.

    Decision: SLA Risk

    Condition example:

    - If Created Date older than 2 days

    Then:

    - (Optional internal flag or message)

11. **Set the final message:**

    Element: Assignment

    - Label: Set Final Message

    Set varActionMessage:

    - High: "High priority ticket detected. Assigned to senior agent."
    - Medium: "Ticket marked as medium priority. Will be handled shortly."
    - Low: "Tickets are low priority and queued for processing."

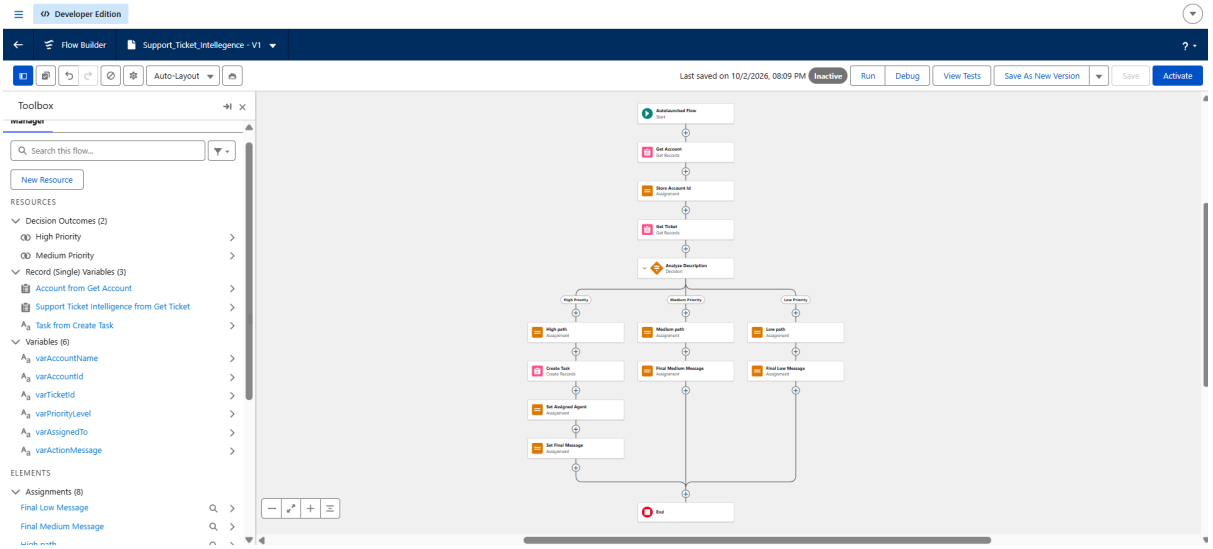
12. **Save Flow:**

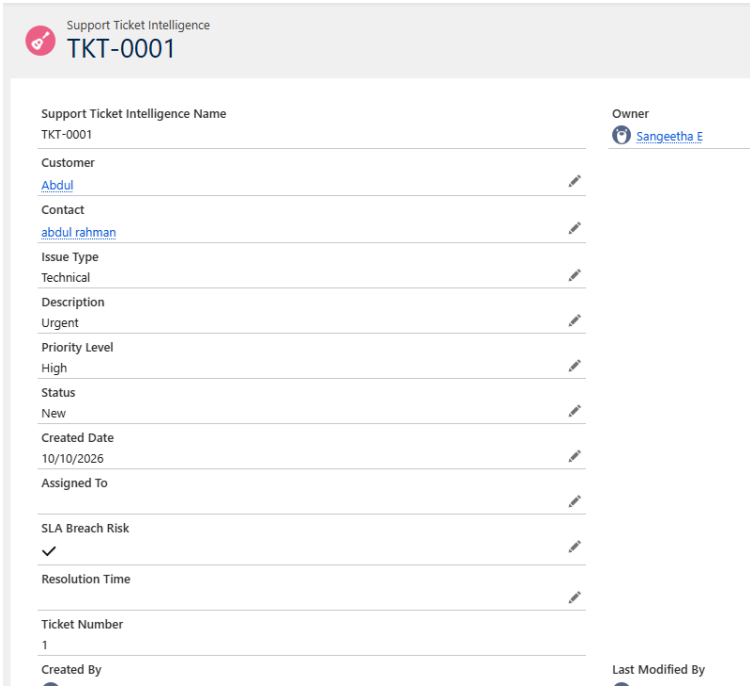
Label Name : Support_Ticket_Intelligence

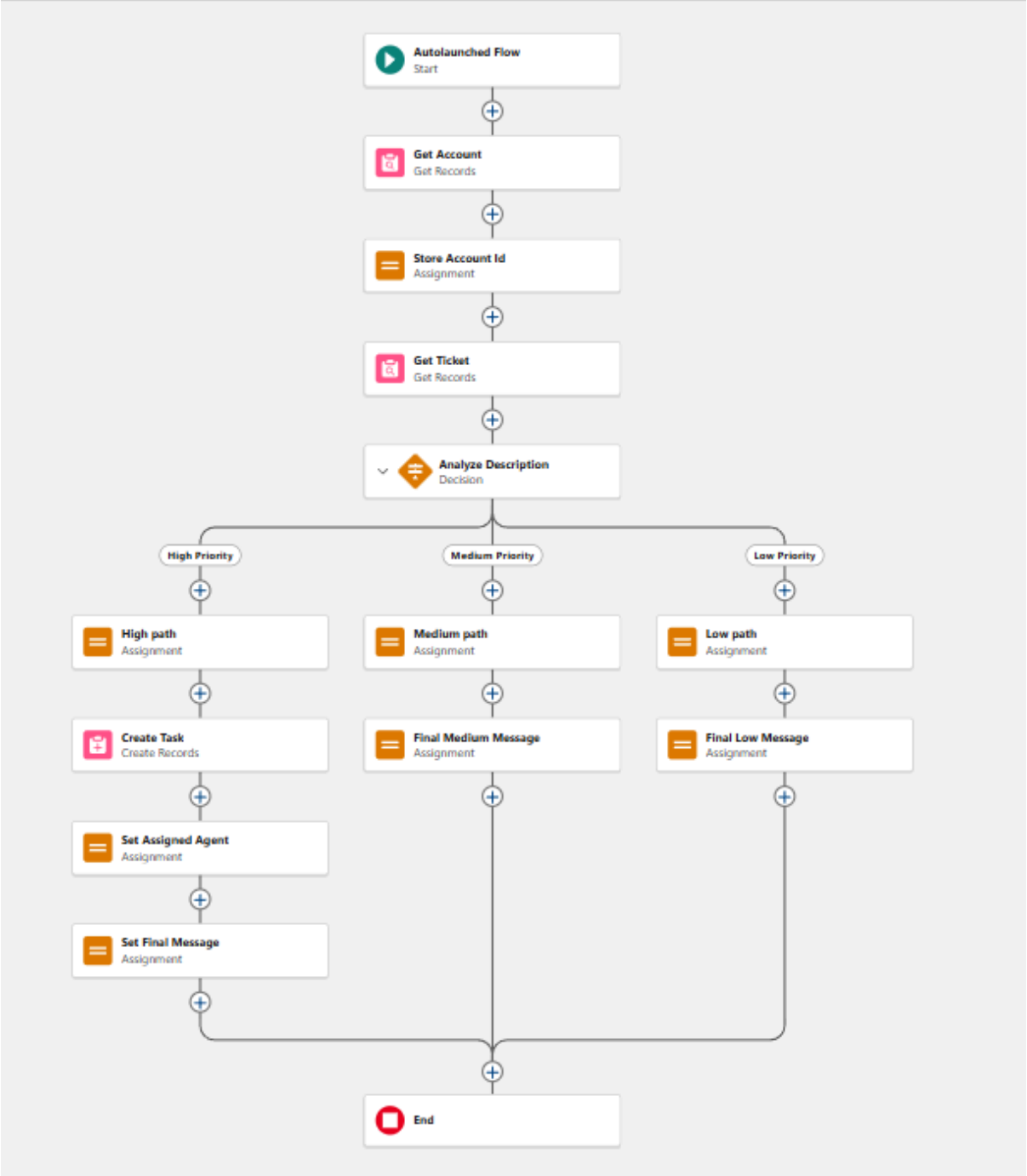
Flow structure

Start → Get Account → Store Account Id → Get Ticket → Store Ticket Id → Analyze Description → Set Priority → Check High Priority → Create Task if High → Set Assignment Output → Optional SLA check → Set Final Message → End

OUTPUT FLOW :







6.3 AGENTFORCE CONFIGURATION

The project uses an Agentforce subagent configured as follows:

- **Name:** Support Ticket Priority Analysis

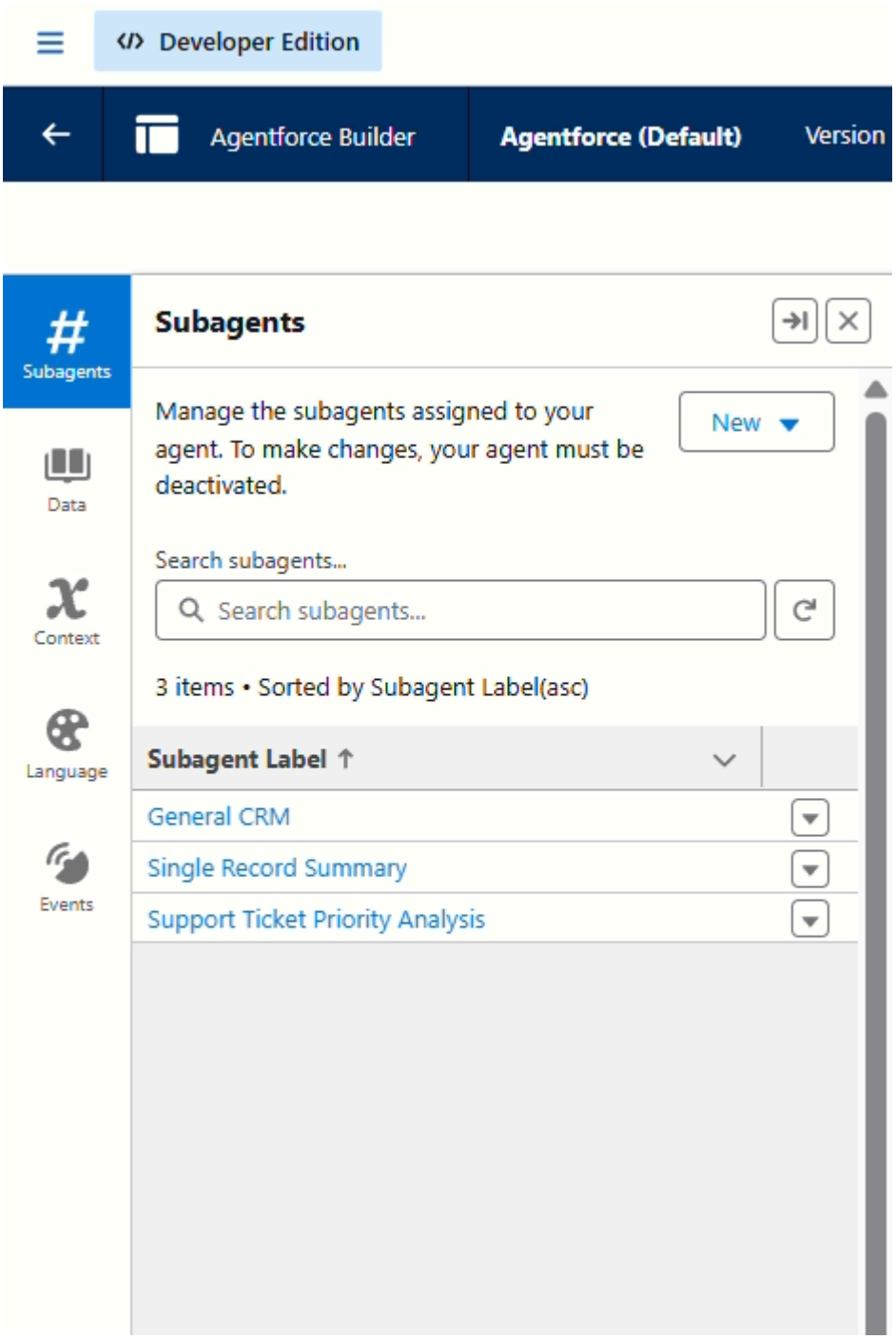
  API name: Support_Ticket_Priority_Analysis
- **Classification description:** This topic analyzes customer support tickets to determine priority based on issue description and urgency. It handles user requests related to ticket status, issue severity, and support prioritization.
- **Scope:** Your job is only to analyze support ticket descriptions, determine the priority level (High, Medium, Low), and trigger backend automation for task assignment. Do

not handle unrelated requests such as billing, subscription management, or account updates.

Subagent instructions

Follow these steps when this topic is triggered:

1. Ask the user to provide the Account Name if not already given.
2. Retrieve the latest support ticket associated with the Account.
3. Read and analyze the ticket description.
4. Determine priority based on keywords:
    - High Priority: contains "urgent", "not working", "failure"
    - Medium Priority: contains "issue", "slow", "delay"
    - Low Priority: if none of the above
5. Set the priority level accordingly.
6. If the ticket is High Priority:
    - Trigger the Auto-Launched Flow
    - Ensure a task is created for immediate handling
7. Assign the ticket to the appropriate support level (e.g., senior agent for high priority).
8. Return a clear response message to the user:
    - High → Inform urgent handling is initiated
    - Medium → Inform issue will be handled shortly
    - Low → Inform ticket is queued
9. Do not update records directly or send emails. Use Flow for all actions.



Agent action
Description: Analyzes customer support tickets using an Account Name, determines priority from the description, and uses the Flow for the configured High-priority action.

Loading text: “Analyzing ticket details and determining priority...”

INPUT

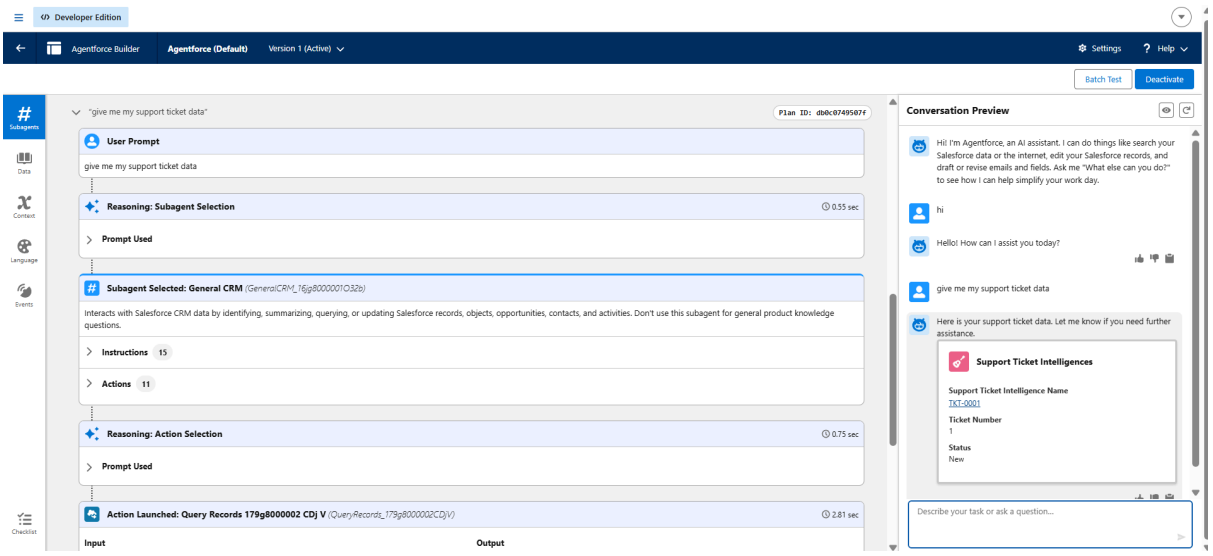
VARIABLE	PURPOSE
varAccountName	Account Name used to retrieve the related ticket

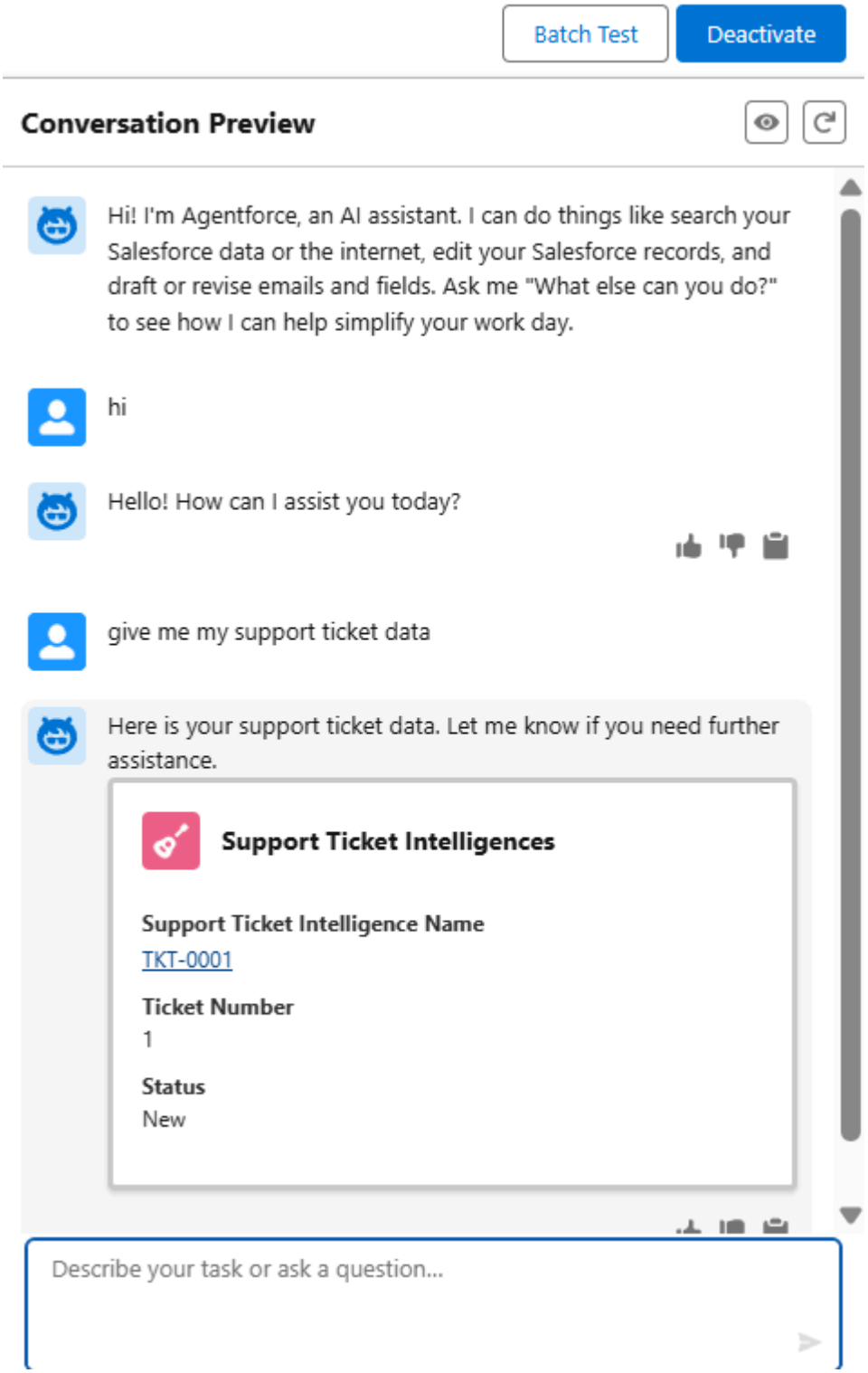
OUTPUT

VARIABLE	PURPOSE
varAccountId	Account ID used for the lookup
varTicketId	ID of the ticket analyzed
varPriorityLevel	Calculated priority: High, Medium, or Low
varAssignedTo	Assignment information returned by the Flow
varActionMessage	Summary of the priority and action

The outputs are configured to be displayed in the conversation as text.

FINAL OUTPUT





7. FUNCTIONAL TESTING

The solution should be tested in Salesforce with sample Accounts and related support tickets. Testing should verify that:

- The Flow finds the Account by the provided Account Name.

- The Flow retrieves the latest related ticket.
- Each keyword group produces the intended priority.
- A High-priority ticket creates a Task with the configured subject, priority, and status.
- The Flow returns the expected output values to Agentforce.
- Agentforce displays the returned ticket and action information.

No specific test results or screenshots were included in the project details provided, so this section describes validation criteria rather than claiming particular test outcomes.

8. ADVANTAGES AND LIMITATIONS

Advantages

- Stores ticket data in a dedicated Salesforce custom object.
- Retrieves the latest related ticket for an Account.
- Applies consistent, configurable keyword-based priority rules.
- Creates an urgent-handling Task for High-priority cases.
- Provides a conversational way to run ticket analysis through Agentforce.
- Returns priority and action information to the user.

Limitations

- Classification is based on a fixed set of keywords and may miss other urgency signals.
- Results depend on accurate Account names and ticket relationships.
- The Flow's assignment output is a text value; the described configuration does not establish that the ticket's Assigned To lookup field is updated.
- The optional SLA check is not confirmed as implemented.
- No automated testing results, performance measurements, or broader assignment rules were provided.
- Flow and Agentforce configuration require maintenance when requirements or keywords change.

9. CONCLUSION

This project combines a Salesforce custom object, an Auto-Launched Flow, and an Agentforce subagent to analyze support tickets. The Flow retrieves the latest ticket associated with an Account and classifies it as High, Medium, or Low using configured keywords. For High-priority tickets, it creates an urgent-handling Task and returns the configured assignment information and action message. The solution demonstrates a conversational approach to ticket analysis and Salesforce automation within the implemented scope.

10. FUTURE SCOPE

Possible extensions include:

- Add more urgency indicators and configurable classification rules.
- Implement and validate the optional SLA-risk check.
- Configure assignment rules that update the ticket's Assigned To field.
- Add error handling for missing Accounts or missing related tickets.
- Add testing and reporting for classification accuracy and Task creation.
- Expand the Flow outputs with additional ticket details, where appropriate.

11. APPENDIX

- **Custom object:** Support_Ticket_Intelligence__c
- **Auto-Launched Flow:** Support_Ticket_Intellegence
- **Agentforce subagent:** Support Ticket Priority Analysis
- **Agent action:** Flow-based support ticket analysis
- **Key related records:** Account, Contact, User, and Task

THANK YOU